

Lecture 1- Part 1:

Professional Practice

in Computing

Marwa Mahmoud

Today's lecture

- Introductions
- Course administration details
- First topic from course material: Professional skills
- Video assessed exercise (AE1)

Introduction to RPS

Course Lecturers

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Course structure

- Lectures on **Fridays** from **1:00-3:00PM** in **JMS438AB**

Self-study:

Pre-lecture readings/ videos are required

Unassessed tutorial / exercises — unassessed (no solutions)

- **Course Assessment:**

- 20% - Video assessed exercise (group assessment)
- 20% - Written assessed exercise (group assessment)
- 60% - Final exam (December)

2 -hour lecture with 10 minutes break

Assessed Exercises

There are 2 assessed exercises:

- Video AE

- Professional skills video assessment (20%) (end of week 4)

- Written AE

- Literature survey/review assessment (20%) (end of week 5)

Both are conducted in teams:

- > Please check your team (group) allocations on Moodle if you haven't
- > Contact me as soon as possible if there is any issue/ concern.

Overview of the course content

Intended Learning Outcomes of Course

By the end of this course students will be able to:

1. Identify the principal social, ethical, legal, and professional issues in information technology (IT);
2. Explain the laws and codes of conduct relevant to the IT industry, and relate them to examples of IT in practice;
3. Constructively criticise the arguments of others;
4. Critically review research papers;
5. Perform a comprehensive literature search and write a comprehensive literature survey;
6. Write an appropriate study design for a chosen CS research problem;
7. Perform critical analysis of experimental designs;
8. Discuss issues relating to the ethics approval of experiments.

Overview of the assessed exercises

There are 2 assessed exercises for this course:

- Video assessed exercise - worth 20%
 - Released: Today
- Written assessed exercise - worth 20%
 - Released: Next week

Video assessed exercise [worth 20%]

- Group exercise (groups of 5 or 4)
- **Groups are already assigned randomly (check Moodle)**

The task : [Create a 90 second video (where the definition of “video” is intentionally vague)]

Why do we need to learn about Professional Practice in Computing ?

Outline

- Computer Misuse Act
- Data Protection laws
- Copyright
- Confidential Information

Computer Misuse Act

Computer Misuse Act - Outline

- Why the Computer Misuse Act came into existence
- What the Computer Misuse Act protects against (generally)
- Awareness of notable or controversial legal cases under the Computer Misuse Act

**Hacking was not
always illegal**

The Prestel Hack (1984–1987)

- Schifreen and Gold were British computer enthusiasts who **accessed British Telecom's Prestel viewdata service** (an early online information system).
- They discovered that the system's administrator password was "1234" — and using it, they gained unauthorised access to private message boxes, including one belonging to the Duke of Edinburgh.
- Although their intent was not malicious (more curiosity and demonstration of weak security), they were charged under the Forgery and Counterfeiting Act (1981) because at that time, no specific law covered hacking.
- They were initially convicted but later **acquitted on appeal** because the law didn't properly apply to computer offences.

Computer Misuse Act 1990

- Covers individuals where the computer, the offender, or both are in the UK
- 3 offences:
 - Unauthorised access to a computer
 - Unauthorised access to a computer with intention to commit a serious crime
 - Unauthorised modification of the contents of a computer

Unauthorised access to computer material (Section 1)

- Accessing any computer, system, or data without permission.
- Example: logging into someone else's account or system without consent.

There was no harm done is not a valid defence

Unauthorised access with intent to commit further offences (Section 2)

- Accessing a system without permission in order to commit another crime, such as fraud or data theft.
- Example: hacking into a bank's system to steal information or money.

Unauthorised modification of computer material (Section 3)

- Where the person lacks permission to modify a computer's content but does modifies it anyway
 - Includes malware, web page redirection, data encryption
 - Done with intention and Knowledge of what is being done

This is the highest level of offence in the Computer Misuse Act

Why don't we hear about more cases?

- Companies want to avoid bad publicity

If you are a company and someone has modified your customers data is it worth the bad publicity if you can quickly revert the data and prevent similar attacks from occurring?

- It requires a lot of police resources to pursue these cases
- Penalties imposed on the guilty have been limited

The Case of Paul Bedworth (1993)

- Bedworth hacked into computers across Europe, including the Financial Times in London, a cancer research institute in Brussels and European Community offices in Luxembourg. The trial judge observed that the full list of computers that had been attacked would be 'as long as a telephone directory'.
- A central plank of Bedworth's defence was the evidence of James Griffith-Edwards, an expert in addictive behaviour, who said that Bedworth had a 'nonchemical dependence' on using a computer. Although he knew what he was doing was wrong, he could not prevent himself – much like an obsessive hand-washer – argued Griffith-Edwards.

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Hacker's acquittal casts doubt on law

By [Charles Arthur](#)

📅 27 March 1993



European action against computer hackers and virus writers is being held back by weaknesses in national laws and the lack of specialised police departments to cope with computer crime. And efforts in the US to prosecute hackers could be hindered by a loophole in the law there which overlooks the increasing miniaturisation of computers.

Britain's police were shocked by last week's acquittal of self-confessed hacker **Paul Bedworth**, who was found not guilty in the first major trial of its kind under the Computer Misuse Act of 1990. Bedworth faced two charges of conspiracy under the act and one of conspiring to dishonestly obtain telecommunications services.

The Case of Paul Bedworth (1993)

- There has been various amendments and new laws introduced to try and address the shortcomings of existing laws/policies
- The problem of technology developing much faster than the legal system

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Summary

- The Computer Misuse Act (1990) was created because existing UK laws didn't cover crimes involving computers.
- Three offences of the act you should know about
- The Computer Misuse Act is far from perfect and amendments and new laws are proposed regularly to try and address its shortcomings

Data Protection

Data Protection/ GDPR – Outline

- Introduce the UK's Data Protection Act
- Terminologies in GDPR and UK's Data protection Act
- Note: a single legislation is never sufficient - this is something which is repeatedly revised

The 1984 Data Protection Act

- The 1st data protection act to protect individuals data against:
 - The use of inaccurate personal information or information that is incomplete or irrelevant
 - The use of personal information by unauthorised persons
 - The use of personal information for purposes other than that for which it was collected

Key terminology

- Personal data: relates to a living person who can be identified by the data
 - Note: this includes expressions of opinion about the person, and intentions towards the person
- Sensitive personal data: relates to ethnicity, sexual orientation, health, political opinions, criminal record, etc

Key terminology

- Data processing: obtaining, recording, holding the data, or carrying out any operations on it (extensive definition of 'operations')
 - It has a very wide definition

Key terminology

- Data subject: the person who is the subject of the personal data
- Data controller: determines how or why personal data is processed
- Data processor: processes personal data on behalf of the data controller

Note: the data processor need not be employed by the data controller but they will need to be in communication with each other in some capacity

The various UK Data Protection Acts

- The 1st data was introduced in 1984
 - The 1984 act was later superseded by the 1998
 - The 1998 act was then superseded by the 2018 act
- Functionally they aim to serve individuals in the same way

What do you need to know about the Data Protection Act?

- The protections it gives individuals
- In this course, we will refer to/ use a condensed summary of the 1998 act
- These protections are also present in the in 2018 act

If you want to read the latest version: <https://www.gov.uk/data-protection>

1998 Data Protection Act (in brief)

1. Personal data shall be processed fairly and lawfully, and not without the subject's consent

Individuals should know how their data is being used.

1998 Data Protection Act (in brief)

1. Personal data shall be processed fairly and lawfully, and not without the subject's consent
2. Personal data shall only be processed for specified and lawful purposes (and not for any other)

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3. Personal data should be adequate, relevant and not excessive in relation to the purpose

Only collect the **minimum amount of data necessary** for the stated purpose.

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2. Personal data shall only be processed for specified and lawful purposes (and not for any other)
3. Personal data should be adequate, relevant and not excessive in relation to the purpose
4. Personal data must be accurate and up to date

Inaccurate data should be corrected or deleted.

1998 Data Protection Act (in brief)

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3. Personal data should be adequate, relevant and not excessive in relation to the purpose
4. Personal data must be accurate and up to date
5. Personal data should not be kept for longer than is necessary for purpose

Data should be **kept only as long as necessary** for the purpose it was collected.

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6. Personal data must take into account the rights of subjects (see later..)

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7. Technical and organisational operations must ensure there is no unlawful processing or loss of personal data

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7. Technical and organisational operations must ensure there is no unlawful processing or loss of personal data
8. Personal data shall not be transferred to a country outside the EU unless it has similar data protection laws

The rights of data subjects

- Data subjects have the right to receive:
 - A description of the personal data being held about them
 - An explanation of the purpose for which it is being held and processed
 - A description of the people or organisations to which it may be disclosed
 - An intelligible statement of the specific data held about them
 - A description of the source of the data

The rights of data subjects

- Data subjects have the right to receive:
 - Prevent processing likely to cause damage and distress
 - Prevent processing for the purposes of direct marketing
 - Claim compensation in the case of damage caused by illegal processing of personal data

Why introduce GDPR?

- The need for new laws!
- Across the various “Legal Issues” slides there’s this theme of a continual need to revise, amend and introduce new legislation
 - The UK had a 1998 Data Protection Act
 - But then it needed GDPR and a 2018 Data Protection Act
- The ever evolving nature of technology results in the need for continual revision and introduction of new laws

Why introduce GDPR?

Increased geographic scope: data subjects in the EU, regardless of where the processing happens

Increased penalties for breach: 4% of annual global turnover or £200million. Apply to both controllers and processors (includes cloud processors)

Consent: must be opt-in, easy to withdraw, 'intelligible'

Breach notification: within 72hours, including notification of customers

Expanded right of access: where and for what purpose data is being processed; controller must provide copy of personal data free of charge

Right to be forgotten: subject can request removal off all data (withdrawal of consent)

Portability: subject can get personal data from one controller and give it to another

Privacy by design: data protection must be considered at the onset of system design, not as an afterthought

Data Protection Officer: must have resources, report to highest level, must be qualified etc.

But GDPR is not perfect!

- GDPR requires a “lawful basis” for processing personal data, typically through seeking consent, or determining “legitimate interest” in processing.

Consider the example of an AR headset: A device equipped with sensors capable of capturing extensive amounts of the user’s surrounding reality

Many of the “risky” activities are not ruled out by GDPR but rather would have to be legally justified through e.g. garnering user consent

- E.g. **the capture and manipulation of a bystander’s image**

These are technologies which will greatly help people

- An AR device can provide real-time navigational information to a blind user, e.g. tell them how far away someone is, if they are walking towards them, what they look like, what they are wearing, their expression, who they are (if recognised by the device), etc

Access to information: What we find is that people are comfortable with some users having access to more information than others:

- Tensions regarding who should have access to what information
- Does the law need to be reflective of this as well?

Source: <https://arxiv.org/pdf/2003.12122.pdf>

Pedestrian Detection with Wearable Cameras for the Blind: A Two-way Perspective

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ABSTRACT

Blind people have limited access to information about their surroundings, which is important for ensuring one's safety, managing social interactions, and identifying approaching pedestrians. With advances in computer vision, wearable cameras can provide equitable access to such information. However, the always-on nature of these assistive technologies poses privacy concerns for parties that may get recorded. We explore this tension from both perspectives, those of sighted passersby and blind users, taking into account camera visibility, in-person versus remote experience, and extracted visual information. We conduct two studies: an online survey with MTurkers (N=206) and an in-person experience study between pairs of blind (N=10) and sighted (N=40) participants, where blind participants wear a working prototype for pedestrian detection and pass by sighted participants. Our results suggest that both of the perspectives of users and bystanders and the several factors mentioned above need to be carefully considered to mitigate potential social tensions.

Author Keywords

wearable camera, accessibility, social acceptance, pedestrian detection, face recognition, crowdsourcing

INTRODUCTION

With advances in machine learning and computing devices that are small and light enough to be worn on one's body, wearable cameras hold the promise to increase real-world accessibility for people with visual impairments (e.g. [5, 32, 36, 42]) with few ideas making it to commercial products such as OrCam [1], Aima [4] and eSight [11]. Given that these devices can capture information in the surroundings that people are uncomfortable sharing or that could be used for surveillance, public use of wearable cameras is conditional on societal consent. Thus, beyond technical challenges, such as battery life, accuracy, form factor, and split-attention, issues such as social acceptability and privacy concerns [26] hold back their use for more independent living.

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Figure 1: Video stimuli shown to our sighted participants during the online and in-situ studies. Both perspectives were included in the online study; in-situ only the user perspective.

With this paper, we aim to contribute to the ongoing discussion on the social tensions with wearable cameras that have sensing capabilities for accessibility (e.g. [3, 13, 14]). Specifically, we focus on pedestrian detection with wearable cameras for blind users. While this technology can provide equitable access to approaching pedestrians with implications for ensuring one's safety [7] or managing social interactions [20], its always-on nature raises privacy concerns for parties that may or may not realize that they are being recorded and/or visually interpreted. The main contribution of this work is that we explore this tension from both perspectives, those of sighted passersby and blind users, taking into account camera visibility and remote versus in-person experience with a working prototype; dimensions lend an opportunity for further insights into prior findings from Profitt et al. [26] and Ahmed et al. [3]. More specifically, our first study is a balanced online 2x2 factorial-design experiment. Participants (N=206), recruited through Amazon Mechanical Turk, responded to technology attitude and other questions before and after being exposed to videos portraying passerby and user perspectives. Videos varied in terms of the wearable camera (visible head-mounted GoPro versus 'invisible

**10 minutes
break!**